Alexis Deighton MacIntyre

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Citizenship: Canadian
Updated: November 10, 2026

Education

2022 **PhD in Cognitive Neuroscience**
University College London – London, United Kingdom
Thesis: The analysis of breathing and rhythm in speech.
Supervisors: Professor Sophie Scott, Professor Antonia Hamilton.

2017 **MPhil in Music & Science** *With distinction*
Wolfson College, University of Cambridge – Cambridge, United Kingdom
Thesis: Rhythm and Metre Modulate Serial Reaction Time Task Learning and Performance. Supervisor: Professor Ian Cross.

2012 **Bachelor of Music** *GPA: 4.29/4.33*
Vancouver Island University – Nanaimo, Canada.

Employment

Oct. 2023 – July 2027 **Leverhulme and Isaac Newton Trust Early Career Fellow**
MRC Cognition and Brain Sciences Unit, University of Cambridge.

Parental leave: May 2026 – April 2027

Oct. 2021 – Sept. 2023 **Postdoctoral Research Associate, Deep Hearing Lab**
MRC Cognition and Brain Sciences Unit, University of Cambridge.

Journal publications

2026 **MacIntyre, A. D.**, Gaultier, C., Goehring, T. Decoding of Speech Acoustics from EEG: Going Beyond the Amplitude Envelope. *Journal of Neural Engineering, 23(1)*, 016014.

2025 **MacIntyre A. D.** The map and the territory: Why diversity advances neuroscience. *Brain and Neuroscience Advances*, 2025:9.

2024 Akter, R. and **MacIntyre, A. D. *** (2024).
Cross-Linguistic Effects of the Speech-to-Song Illusion in Speakers of Bangla and English. *Quarterly Journal of Experimental Psychology*, 17470218241293627. ** Senior author.*

MacIntyre, A. D., Carlyon, R. P., and Goehring, T. (2024).
Neural decoding of the speech envelope: Effects of intelligibility and spectral degradation. *Trends in Hearing*, 28.

2023 **MacIntyre, A. D.**, Lo, H. L. J., Cross, I. and Scott, S. K. (2023).
Task-Irrelevant Auditory Metre Shapes Visuomotor Sequential Learning. *Psychological Research*, 1-22.

2022 **MacIntyre, A. D.**, and Scott, S. K. (2022).
Listeners are sensitive to the speech breathing time series: Evidence from a gap detection task. *Cognition*, 225, 105171.

MacIntyre, A. D., Cai, C. Q. and Scott, S. K. (2022).
Pushing the envelope: Evaluating speech rhythm with different envelope extraction techniques. *The Journal of the Acoustical Society of America*, 151(3), 2002-2026.

2018 **MacIntyre, A. D.** (2018). The signification of the signed voice. *Journal of Interdisciplinary Voice Studies*, 3(2), 167-183.

Conference proceedings

2026 Pessanha, F., de Wildt, G., **MacIntyre, A.D.**, Kaya, H., Akdag, A. (2026). Breathing Signal Prediction from Speech: Toward Reproducible and Comparable Models. *ICMI '26: Proceedings of the 28th ACM International Conference on Multimodal Interaction*, Napoli, Italy.

2024 Y. E. Belghiti, K. Daoudi, K. Minaoui and **A. D. MacIntyre** (2024). Automatic detection of speech inspiration using SVM and decision trees. *IEEE 12th International Symposium on Signal, Image, Video and Communications (ISIVC)*, Marrakech, Morocco.

2023 **MacIntyre, A. D.** and Goehring, T. (2023). Effects of spectral degradation on the cortical tracking of the speech envelope. *Proceedings of INTERSPEECH*, Dublin, Ireland, ISCA.

MacIntyre, A. D. and Werner, R. (2023). An Automatic Method of Speech Breathing Annotation. *Proceedings of the 34th Conference on Electronic Speech Signal Processing (ESSV)*, Munich, DE.

2020 **MacIntyre, A. D.**, Rizos, G., Batliner, A., Baird, A., Amiriparian, S., Hamilton, A. and Schuller, B. W. (2020). Deep attentive end-to-end continuous breath sensing from speech. *Proceedings of INTERSPEECH*, Shanghai, China, ISCA.

Schuller, B. W., Batliner, A., Bergler, C., Messner, E. M., Hamilton, A., Amiriparian, S., ... **MacIntyre, A. D.** and Hantke, S. (2020). The interspeech 2020 computational paralinguistics challenge: Elderly emotion, breathing and masks. *Proceedings of INTERSPEECH*, Shanghai, China, ISCA.

Book chapters

2026 **MacIntyre, A. D.** (2026). Speech Rhythm in Hearing Loss and Cochlear Implant Listening. In L. Meyer & A. Strauss (Eds.), *Rhythms of Speech and Language: Physiology, Cognition, Culture* (pp. 743–765). Cambridge: Cambridge University Press.

Preprints and manuscripts in preparation

MacIntyre, A. D., Goehring, T., and Davis, M. (2026). Flexible neural encoding predicts the comprehension of degraded speech. Available from bioRxiv. *Under revision, Imaging Neuroscience.*

MacIntyre, A. D.* and Shahidi, L. (In preparation). Neural representation of human and AI-generated voice-cloned speech. * *Joint first author.*

MacIntyre, A. D.* and Shahidi, L. (In preparation). Enhanced comprehension of reverberant AI-generated speech. * *Joint first author.*

Awards, scholarships, and recognition

2026 **Featured Talk**
Virtual Conference on Computational Audiology (VCCA 2026)

Post-Doctoral Travel Award
Association for Research in Otolaryngology 49th Annual MidWinter Meeting, Puerto Rico
$750

2024 **Scholarship Award**
7th International Conference on Cognitive Hearing Science for Communication, Linköping, Sweden
Conference registration, accommodation, and travel

2021 **1st Place Prize in Mind and Behaviour**
Graduate School Poster Competition, University College London
£400

2019 **Early Achievement Distinguished Alumni Award**
Vancouver Island University

2018 **The Brain Prize and FENS stipend Award**
The Brain Conferences: Computational Neuroscience of Prediction, Rungstedgaard, Denmark
€1,000

2017 **Overseas and Graduate Research Scholarships**
Now known as the Research Excellence Scholarship
University College London
Full PhD tuition, research costs, and maintenance

2016 **Canada Cambridge Scholarship**
University of Cambridge
One awarded nationally
Full MPhil tuition and maintenance

2016 **Merit Scholarship**
University of Amsterdam
€15,000, *declined*

Grants and funding

2024 Accelerate-C2D3 Small Grant: Novel Applications of AI for Research and Innovation, University of Cambridge
£27,721 *Co-PI with Dr. Lidea Shahidi*

Cambridge Language Sciences Workshop Fund Grant, University of Cambridge
£2,000 *Co-applicant with Dr. Jérémy Giroud*

Public Engagement Starter Fund, University of Cambridge
£1,975 *Co-applicant with Dr. Lidea Shahidi*

Conference Travel Grant, Gordon Research Conferences
$1,000 USD

2023–2026 Junior Research Fellowship, Wolfson College, University of Cambridge
£2,400 in research expenses

Leverhulme and Isaac Newton Trust Early Career Fellowship
$< 20\%$ *Success rate nationally*
Full salary and overhead costs, plus £18,000 in research expenses

British Academy Postdoctoral Fellowship
$< 20\%$ *Success rate nationally*
£324,867, *declined*

2023 Guarantors of Brain Small Meeting Grant
Co-applicant with Dr. Abbie Bradshaw
£3,000

2020 INTERSPEECH2020 ISCA Student Grant
Full conference registration

2017 Grindley Travel Grant, Experimental Psychology Society
£500

Music Faculty Travel Grant
£300

Invited talks

July 2027 *Upcoming:* Decoding Speech Acoustics from EEG
Conference on Implantable Auditory Prostheses (CIAP), Banff, Alberta

Apr. 2027 *Upcoming:* Adaptive neural encoding predicts individual speech comprehension
Institute of Cognitive Neuroscience, University College London, UK

Apr. 2025 The Cognitive Neuroscience of Speech Perception
Padjadjaran University, Bandung, Indonesia

Feb. 2025 Speech Science Forum
Psychology and Language Sciences, University College London, London, UK

May 2024 The Neuroscience of Speech Perception
Downing College Psychological and Behavioural Sciences Society, University of Cambridge

Dec. 2023 Decoding Speech from the Brain: Effects of Intelligibility and Spectral Degradation
Brain and Intelligence Science Alliance Young Scientist Forum, Fudan University, Shanghai, China

Jan. 2019 Respiration and the Flow of Language
Linguistics Department, University of Kent, UK

Dec. 2018 The Analysis of Speech Rhythm
Psychology Department, Bangor University, UK

Conference presentations

June 2026 Decoding of speech acoustics from EEG: going beyond the amplitude envelope. *The Virtual Conference on Computational Audiology (VCCA2026), online.*

Feb. 2026 Neural Encoding of Acoustic and Linguistic Speech Features Reveals Correlates of Understanding. *Association for Research in Otolaryngology 49th Annual MidWinter Meeting, Puerto Rico*

June 2024 Neural Decoding of the Speech Signal with Applications to Cochlear Implants. *7th International Conference on Cognitive Hearing Science for Communication, Linköping, Sweden*

Sept. 2023 Speech perception under acoustically degraded listening conditions: The Role of Language Familiarity. *Annual Scientific Meeting of the British Association for Cognitive Neuroscience, Cardiff, Wales*

June 2023 Cortical tracking of speech: Effects of intelligibility and spectral degradation. *The Virtual Conference on Computational Audiology (VCCA2023), online.*

Mar. 2023 An Automatic Method of Speech Breathing Annotation. *34th Conference on Electronic Speech Signal Processing (ESSV), Munich, DE.*

June 2022 Developing Automatic Methods for the Detection of Speech Breathing Inhalations from Breath Belt Data
Respi II Speech Respiration Workshop, Saarland University, DE (online)

Feb. 2022 Pushing the envelope: Evaluating speech rhythm with different envelope extraction techniques
DGfS 2022 (Annual Meeting of the German Linguistic Society), University of Tübingen, DE (online)

Oct. 2020 Deep Attentive End-to-End Continuous Breath Sensing from Speech
INTERSPEECH2020, Shanghai, China (online)

Apr. 2018 Metric Structure Shapes Early Implicit Learning
The Brain Conferences: Computational Neuroscience of Prediction, Rungstedgaard, Denmark

Posters

Mar. 2026	Neural Encoding of Acoustic and Linguistic Speech Features Reveals Correlates of Understanding. *Cognitive Neuroscience Society 2026 Annual Meeting, Vancouver, Canada*
Sept. 2025	Tracking of Acoustic and Linguistic Speech Features by the Brain *8th International Conference on Auditory Cortex, Maastricht, NL*
Sept. 2024	Neural decoding of speech: It goes beyond the envelope *Auditory Science Meeting, Cambridge, UK*
May 2024	Temporal Processing of Speech-Related Breathing *Gordon Research Seminar on Neural Mechanisms of Acoustic Communication, Newry, Maine, USA*
Sept. 2023	Cortical Tracking of Spectrally Degraded Speech *Basic Auditory Science Meeting, London, UK*
July 2023	Cortical Tracking in Speech Perception with Applications to Cochlear Implants *Conference on Implantable Auditory Prostheses, Lake Tahoe, California*
Apr. 2023	Cortical Tracking in Speech Perception with Applications to Cochlear Implants *British Cochlear Implant Group Meeting, Cambridge, UK*
Sept. 2022	Cortical Tracking in Speech Perception with Applications to Cochlear Implants *Hearing, Audio and Audiology Sciences Meeting, Southampton, UK*
May 2022	Listeners Are Sensitive to the Speech Breathing Time Series: Evidence from a Gap Detection Task *Annual Meeting of the British Association for Cognitive Neuroscience, Birmingham, UK*
June 2021	Listeners are sensitive to violations of natural speech breathing timing *Rhythm Production and Perception Workshop 2021, Oslo, Norway (online)*
	Task irrelevant auditory metre shapes visual-motor sequential learning *Rhythm Production and Perception Workshop 2021, Oslo, Norway (online)*
Aug. 2019	"Mean Speech Rate" Doesn't Mean Much: Analysis of Speech Rhythm Benefits from Quantising Inter-Onset Intervals *Eleventh Annual Meeting of the Society for the Neurobiology of Language, Helsinki, Finland*
July 2019	The Flow of Language: Beat-Based Timing Enhances Synchronisation of Speech and Speech Breathing in a Joint Speaking Task *Joint Action Meeting (JAM VIII), Genoa, Italy*

Apr. 2018 The Rhythmic Coordination of Breathing During Joint Speech and Imagined Joint Speech
Annual Research in Imagery and Observation Group Meeting, Bielefeld, Germany

Public communication and media

2022 **Beyond the Voice Box: How Ontologies of the Voice Determine Who, and How, We Sense**
Luminomelia Philosophical Society, Cambridge, UK

2020 **The voice in ancient Greek and Roman society**
Voice Box: An Evening Of Comedy About Voices, Identity And Experience, London, UK

MacIntyre, A. D. (Mar., 2019). What is a Voice? *Sounding Out!* ISSN 2333-0309.

Teaching experience

Oct. 2024 **Guest Lecturer, *Graduate Seminar, Psychology Department, Padjadjaran University, Bandung, Indonesia***
Gave a one-hour introductory lecture on the neuroscience of speech and language to graduate students.

Spring 2021 **Seminar Leader and Examiner, *Quantitative Methods and Mathematical Thinking module*, University College London**
Taught section of undergraduate course on quantitative tools for exploring and analysing data, including introduction to statistics and computer programming in python.

Examiner, *Knowledge, Learning and Inference module*, University College London
Computational formalism and methods in cognitive science, including Bayesian methods, symbolic approaches from artificial intelligence, machine learning techniques, and neural networks.

Examiner, *Psychology, Memory and Decision-Making module*, University College London
Fundamental topics in psychology, such as the comparison of memory systems and heuristics in decision-making.

Supervision and External Evaluation

2026 **Nashao Zhong, MPhil Cognitive Neuroscience, Cambridge University**
Project: Neural correlates of synthetic speech perception

Rain Ng, MPhil Foundations of Clinical Psychology, Cambridge University
Project: Neural correlates of synthetic speech perception

2025 **Susie Black, Psychological and Behavioural Sciences Part II Project, Cambridge University**
Project: Decoding speech perception from the brain

Emilee Gosnell, NIHR Doctoral Clinical Academic Fellow, Cambridge University
Upgrade examiner for PhD in Clinical Neurosciences

2020 – 2021 **Rakhi Akter, Applied Medical Sciences Student, UCL**
Project: The speech-to-song illusion in Bengali and English speakers

2020 – 2021 **Jakab Pilaszanovich, MSc Student, UCL**
Project: Auditory Not-Just-Right Experiences: Rhythm violation as a source of sensory imperfection

2019 – 2020 **Josephine Lo, Laidlaw Scholar, UCL/The University of Hong Kong**
Project: Does rhythm affect how we learn?

Spring 2019 **Bangjie Wang, Affiliate Psychology Student, UCL/Grinnell College**
Project: Classifying rhythmic speech patterns in Mandarin and English

Julia Porter, Affiliate Psychology Student, UCL/McGill University
Project: The salience of vowels as a classification feature in spontaneous versus controlled speech

Workshops and tutorials given

2023 **Plethysmography**
COGNESTIC Summer School, MRC Cognition and Brain Sciences Unit, University of Cambridge

2022 **The Measure and Analysis of Respiratory Effort**
Methods Day, MRC Cognition and Brain Sciences Unit, University of Cambridge

Research roles

Sept. 2014 – Aug. 2016 **Research Assistant, Music and Neuroscience Lab**
Mentors: Professor Jessica Grahn (Western University).

Reviewing

Ear and Hearing, International Society of Music Information Retrieval, Journal of the Acoustical Society of America, Journal of Interdisciplinary Voice Studies, Frontiers in Neuroscience, Journal of Phonetics, Philosophical Transactions of the Royal Society B, Trends in Hearing

Service

April 2026 **Cambridgeshire Deaf Association (Cambridge, UK)**
Co-organised and facilitated a participatory workshop for people with lived experience of, or affected by, hearing loss, gathering community input on research priorities. With Dr. Lidea Shahidi. Funded by the University of Cambridge Public Engagement Starter Fund.

2025 **In2science, UK**
In2research Mentor supporting young people from disadvantaged backgrounds with the broad goal to boost diversity and inclusion in STEM.

2024 – 2025 **Interdisciplinary Voice: From Motor Control to Identity**
Co-founder of a recurring small meeting at the interface of brain and behavioural sciences, linguistics, speech technology, and other areas relating to the voice. With Dr. Abbie Bradshaw. Funded by the Guarantors of Brain Small Meeting Grant.

2023 – 2024 **Undergraduate Admissions Panel, Christ's College, University of Cambridge**
Interviewed applicants to Psychological and Behavioural Sciences, and scored and ranked candidates on interview performance and written application.

2023 **Supervisor, SHARE (Social Sciences & Humanities Access to Research Experience), University of Cambridge**
Summer project placement to increase access to research for students who are currently underrepresented in postgraduate research

2022–Present **Equality and Diversity Committee, MRC CBSU**
Promoting and improving departmental ED initiatives, planning annual Festival of Diversity event, and steering committee member and project demonstrator for the MRC CBSU's BrainBus programme for widening access.

2021 **IMPRS NeuroCom Summer School, Online (London, UK)**
Co-organised the 10th IMPRS NeuroCom Summer School in Cognitive Neuroscience, a joint virtual event for students and ECR hosted by the Institute of Cognitive Neuroscience at UCL, UK and IMPRS NeuroCom at MPI CBS and Leipzig University

2019 **Big Bang Science Event at the City Academy Hackney (London, UK)**
Designed and presented STEM workshops to year 8, 9, and 10 science students.

2019 – 2020 **Reach Out, London, UK**
Worked with young people from schools in disadvantaged/over-exploited communities to build life skills and improve academic performance.
Awarded "Mentor of the Term"

2016 – 2017 **Cambridge Singing for Breathing, Cambridge, UK**
Helped run therapeutic singing and exercise sessions funded by the British Lung Foundation for people with chronic lung conditions in the Cambridgeshire area

2014 – 2016 **Guelph Black Heritage Society, Guelph, Canada**
Planning committee for an annual series of community events honoring Black contributions to Canadian society

Society memberships

British Association for Cognitive Neuroscience, British Neuroscience Association, British Cochlear Implant Group, Society for the Neurobiology of Language, Timing Research Forum, Association for Research in Otolaryngology, Cognitive Neuroscience Society, Association for Computing Machinery

Technical expertise

Programming
MATLAB, python, R, HTML/CSS, LaTeX, Git

Software
SLURM Workload Manager, Adobe CS, Audacity, PRAAT, PsychoPy, ProtoKinetics Movement Analysis Software, ActiView, labVIEW, LabChart

Research
Electroencephalography, gait recording and analysis, respiratory inductance plethysmography

Languages
English (primary language), Dutch (low-intermediate), French (beginner), German (beginner)